STUDENTS' TECHNICAL COUNCIL



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TECHSHILA

INTER BHAWAN TECH MEET

Computer Vision

The Low-Light Image Restoration Challenge

The Low-Light Image Restoration Challenge aims to improve the quality of images captured in low-light conditions, making them comparable to those captured in normal-light. With the growing demand for high-quality images and videos, this challenge is of immense importance in various fields, including surveillance, photography, and medical imaging. A novel method for low-light image restoration would have far-reaching implications, including improved safety, better diagnosis, and enhanced visual media.

Problem Statement

Meet Khushal, a student at IIT Roorkee who loves photography. Khushal is an avid photographer and often spends his free time capturing photos of the beautiful campus and its surroundings. However, Khushal has noticed that his low-light images never turn out as good as he wants them to. The images are often too dark, have poor colors, and high noise levels. He has tried using various image editing software, but none of them seem to work.

This is where you come in. Khushal has heard about the low-light image enhancement challenge, and he needs your help. Your task is to build a deep learning model that can convert a low-light image to a high-quality, well-lit image. By doing so, you will be able to help Khushal and other aspiring photographers like him to improve the quality of their low-light images.

Task

Your task is to train a neural network model on the LOL dataset, which contains 500 pairs of low-light and normal-light images. The neural network should take a low-light image as input and produce a normal-light image as output. The model should be able to remove noise, adjust the brightness and contrast, and produce images that look like they were taken in normal-light conditions.

You can use pre-trained models such as ResNet or VGG to build your model. You can also use image enhancement techniques such as histogram equalization, gamma correction, or contrast stretching to improve the quality of the images.

Dataset

The LOL dataset is a collection of 500 low-light and normal-light image pairs. It is divided into 485 training pairs and 15 testing pairs. The low-light images contain noise produced during the photo capture process. The dataset is available at the following link: https://paperswithcode.com/dataset/lol

Evaluation

The performance of the neural network model will be evaluated based on the PSNR (Peak Signal-to-Noise Ratio) metric. During the final evaluation, you will be provided with around 50 unseen images, and your neural network model should give high PSNR on those images. The higher the PSNR, the better the image quality.

The points distribution is as follows:

1. Total Event Points: 100
2. PSNR weightage: 50%
3. Presentation weightage: 30%
4. Latency weightage: 20%

Participation

Participation details related to the challenge:

1. The maximum team size is 6 members.
2. Contact your bhawan's contingent leader to participate.
3. For first-year undergraduates, there can be unlimited teams, and a mentor will be assigned to all teams.

Submission

You need to submit a well-formatted and well-annotated iPython Notebook which should clearly explain your approach, your model, etc. Details regarding the submission will be communicated via discord once the teams have been finalised.

The submission details are as follows:

13th April: Code Submission
14th April: Poster Presentation

Resources & Conclusion

Low-light image enhancement is an exciting field of research that has the potential to revolutionize photography. With the LOL dataset, you can train your neural network models to enhance low-light images and make them look like they were taken in normal-light conditions. Here are some resources to help you out:

- http://openaccess.thecvf.com/content/CVPR2022/papers/Xu_SNR-Aware_Low-Light_Image_Enhancement_CVPR_2022_paper.pdf

We look forward to seeing your submissions and wish you the best of luck!